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Question	Answer	Marks
1	**Defining the problem**	
	Mass of cylinder *m* is the independent variable and period *T* is the dependent variable, or vary mass of cylinder *m* and measure period *T.*	**1**
	Keep radius of cylinder constant.	**1**
	Methods of data collection	
	Labelled diagram of workable experiment including: • beaker with (cooking) oil on a bench or container supported by stand where stand is on a bench • cylinder partially submerged in (cooking) oil • cylinder and (cooking) oil labelled.	**1**
	Method to determine mass *m* of cylinder, e.g. use a (top pan) balance.	**1**
	Method to determine period or *T*, e.g. use a stopwatch / timer to time oscillations.	**1**
	Method to determine diameter of cylinder, e.g. micrometer or calliper	**1**
	Method of Analysis	
	Plots a graph of T^2 against *m.* (Allow other valid graphs, e.g. lg *T* against lg *m*)	**1**
	Relationship valid if a straight line passing through the origin is produced. (Allow gradient = 0.5 for log *T* against log *m*).	**1**
	$K = \frac{4\pi}{\text{gradient} \times \sigma r^2}$ ($K = \frac{4\pi}{10^{2\times\text{y-intercept}} \times \sigma r^2}$ for lg *T* against lg *m*).	**1**

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1	**Additional detail including safety considerations** **Max 6**	**6**
	Use gloves <u>to prevent oil</u> contacting skin / slippery hands OR Perform experiment in a tray <u>to prevent oil spillages</u>.	D1
	Keep density / temperature of the (cooking) <u>oil</u> <u>constant</u> or keep σ <u>constant</u>.	D2
	Mass of oil = mass of beaker and oil – mass of beaker <u>and</u> use a measuring cylinder to determine the volume of the oil. Do not accept (calibrated) beaker.	D3
	Methods to measure volume of oil and determine mass of oil and use equation density σ = mass / volume for measurements.	D4
	Time n oscillations and divide nT by n where $n \geqslant 5$.	D5
	Description of method of counting oscillations with position of fiducial mark / mark on cylinder / beaker / fixed point shown in diagram.	D6
	Repeat experiment for each value of m and average T.	D7
	r = diameter / 2 provided diameter measured.	D8
	Repeat measurements of <u>diameter</u> <u>in different directions</u> and average.	D9
	Wait for oscillations to become even / steady.	D10

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